

Predicting the Big Bash League Player of the Match

A Comparative Analysis of Statistical and Regression Models

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Player of the Match (PotM) award, recognises the player whose contribution fundamentally altered the game's outcome for their team's betterment.

Executive Summary

Can the Big Bash League Player of the Match (PoTM) be predicted using only objective performance data?

The report was motivated by the ambiguous nature of PotM award systems and aims to introduce a more data-driven methodical approach. This report demonstrates that objective performance data can be used to predict the PotM from match groupings. This project analysed 10 seasons of Big Bash League (BBL) data (7-men, 3-women) and applies four different modelling techniques. Player performance is measured using summarised match data and then worked to train machine learning models. The models used were: penalised Logistic Regression (LR) variants, conditional Generalized Linear Model (GLM) variants, a Generalized Additive Model (GAM), and a Random Forest (RF) ensemble.

Three consistent findings emerged from the modelling comparison:

- Fundamental Problem:
 - Candidates for each match total 22 (11 players per team), with only one player receiving the PotM award. This extreme class imbalance means a model that fails to identify any winner can still achieve a 95.45% accuracy score.
- Batting/bowling bias:
 - Given the high-pressure environment of T-20 cricket it is expected that batters take more risk than they would in other formats. Coupled with limited overs (~120 balls per innings, 1 innings per team) batting performance is a highly significant indicator of PotM outcome. In general bowling contribution affects PotM award, but is seen as a secondary driver for batting performance.
- Model Trade-offs:
 - There is an inherent trade-off between models identifying a single winner whilst providing well-calibrated probabilities. The logistic regression (elastic net, global scaling) achieved the highest top 1 accuracy picking the PotM winner in ~69% of matches. However, its calibration performed poorly. A GAM approach yields slightly lower top-1 accuracy (~67%) but produced the most reliable probabilities and the strongest shortlisting

results. The RF ensemble's conservative probability distribution, increasing calibration but reduced accuracy.

Model performance was evaluated under two conditions. First, how well each model ranked the true Player of the Match (PotM), and second, how true probabilities were.

The results lead to the conclusion that there is no best model that fits all criteria. The preferred approach is dependent on operator intent.

This project's timing coincides with cricket's increasing embrace of technology and analytics. Innovations like the Duckworth-Lewis-Stern (DLS) method, Hawk-eye, and UltraEdge enhance objectivity in decision adjudication. In the same spirit, this analysis uses statistical models to reduce systemic bias in player recognition by quantifying performance impact transparently. Whilst using cricket, the PotM prediction method generalises to other team sports where similar performance impact is awarded.

Additional applications include:

- Broadcasting platforms using models to highlight likely PotM candidates during and after matches.
- Coaches and team management setting objective benchmarks for individual contribution. Mapping trends over time.
- Provides a transparent and unbiased framework for player performance. This transparency offers a clear rationale for PotM awards.
- Sporting bookmakers use of probabilities manages log-odds distribution. Calibrated probability estimates translate to improved risk management.

In summary, the findings demonstrate a viable, tested baseline model for PotM prediction that complements human judgement and supports fair recognition of player performance.

Section 1: Introduction

The T-20 Big Bash League (BBL) is Australia's dominant cricket competition, where displays of merit can dramatically shift match outcomes. Lacking a formal criterion for PotM I offer, formally recognising a player whose contribution fundamentally altered the game's outcome for their team's betterment. Similar PotM award structures exist throughout sport, yet the selection process is ambiguous, often blending objective statistics with influence-bias. Fans and commentators on occasion debate these choices, insighting that unweighted factors like a spectacular moment or popular fan sentiment overwhelm this decision making. This raises the intriguing question: ***Can we predict the PotM using only objective performance data?***

The question poses a discrete choice problem: in each match, out of 22 players, exactly one is chosen. The goal is not to perfectly replicate historical judgement but to discover the data-defined patterns that underlie PotM selection.

Cricket as a sport has shown an increased willingness to use technology and data to inform crucial decision making. The international Cricket Council (ICC) has implemented tools like the Duckworth-Lewis-Stern method used to define fair run targets in rain-restricted conditions, the Hawk-Eye system reviews LBW appeals, and the UltraEdge system for detecting bat-ball contact. These tools show ICC intent to make the game's outcome more objective. In the same manner, PotM modelling provides clarity, by quantifying contribution in a measurable way eliminating bias-controversy while complementing the traditional framework.

Section 2: Exploratory Data Analysis

To explore this idea, 10 seasons of BBL data are analysed (seven men's BBL, three women's WBBL). Developing and comparing four modelling approaches which exploitative leverage vary for data processing techniques.

2.1- Raw Data and Preprocessing:

The dataset for this project was derived from ball-by-ball match record and match summary metadata. By constructing a player-match level dataset singular player performance can be observed. The target variable is a binary flag indicating whether that player was the PotM (1-true, 0-false).

After cleaning the data, the final dataset covers 424 matches spanning the 10 seasons. Of the 424 matches 422 PotM instances were observed (2 matches with no award).

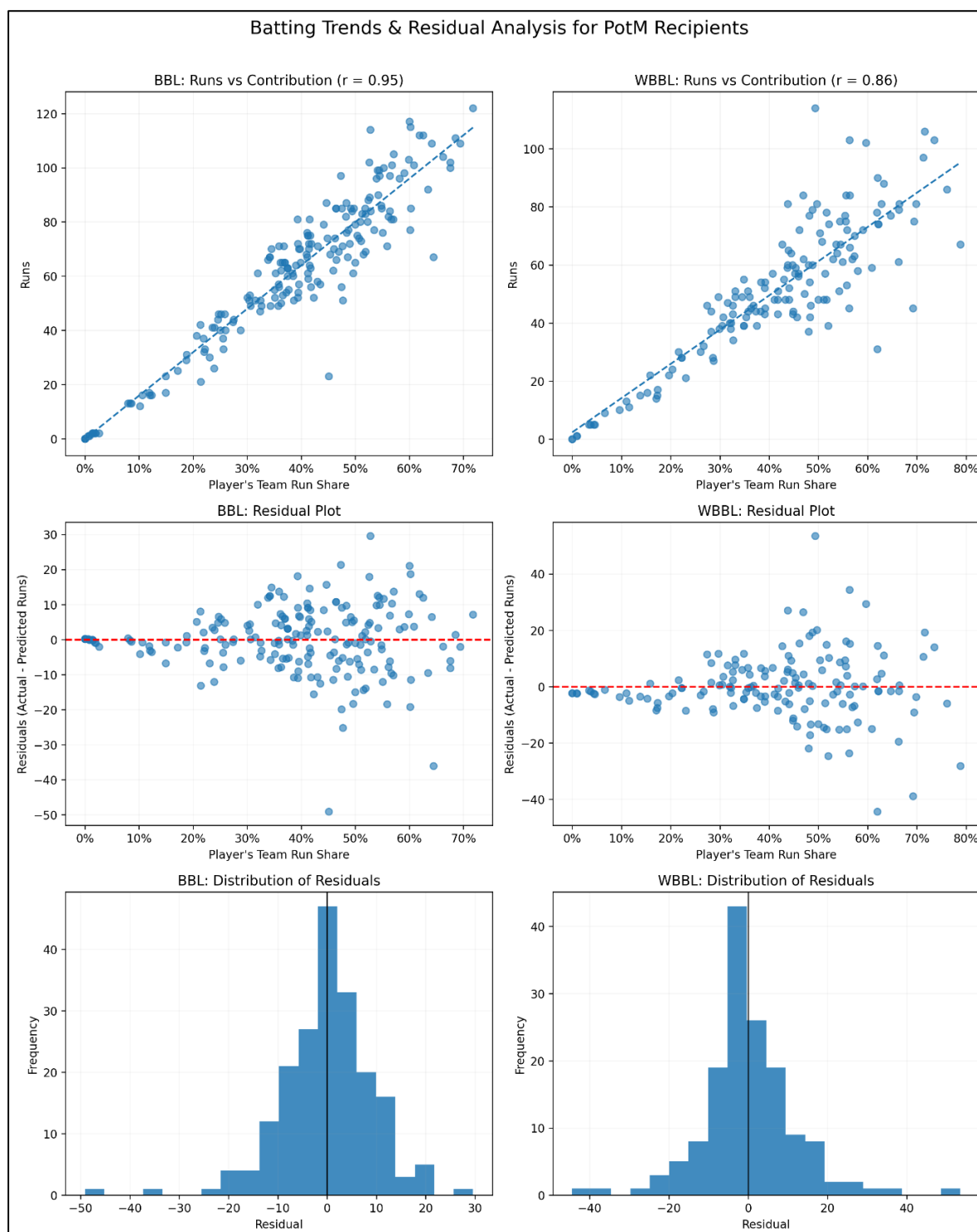
Initial Features:

- Player Performance:
 - Runs scored, Balls Faced, Boundaries (fours, sixes), Strike rate, Overs bowled, Wickets taken, Runs conceded, Dot balls, Bowler Economy,
- Match Information:
 - Season, League, Innings, Winner, Toss result, Home/away designation.
- Player Traits:
 - Categorical flags for Batter, Bowler, All-rounder.

These features form the basis for feature engineering applied for modelling.

2.2- Initial Data Analysis:

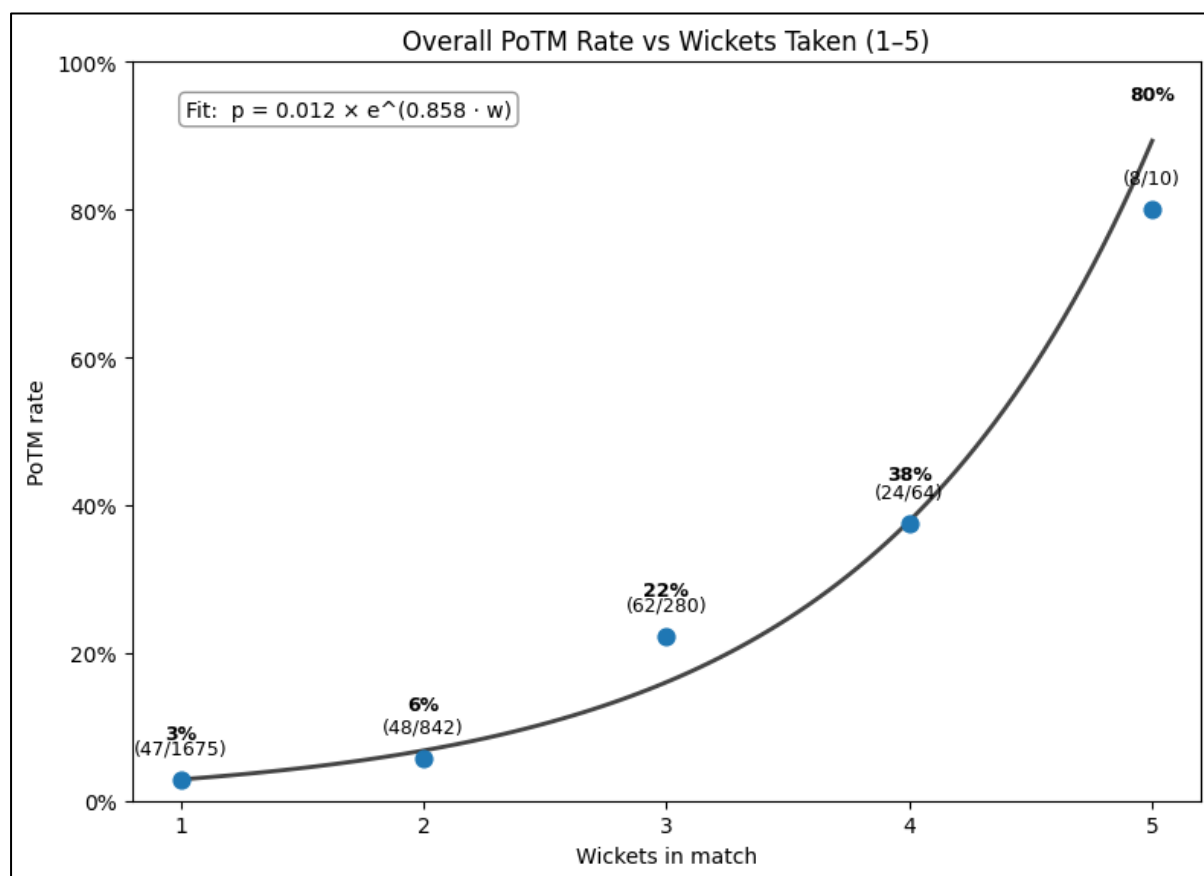
From the initial frame, early insight into the condition under which PoTM awards were made are recorded and examined through descriptive tabulation.

Figure 1: Batting contribution:

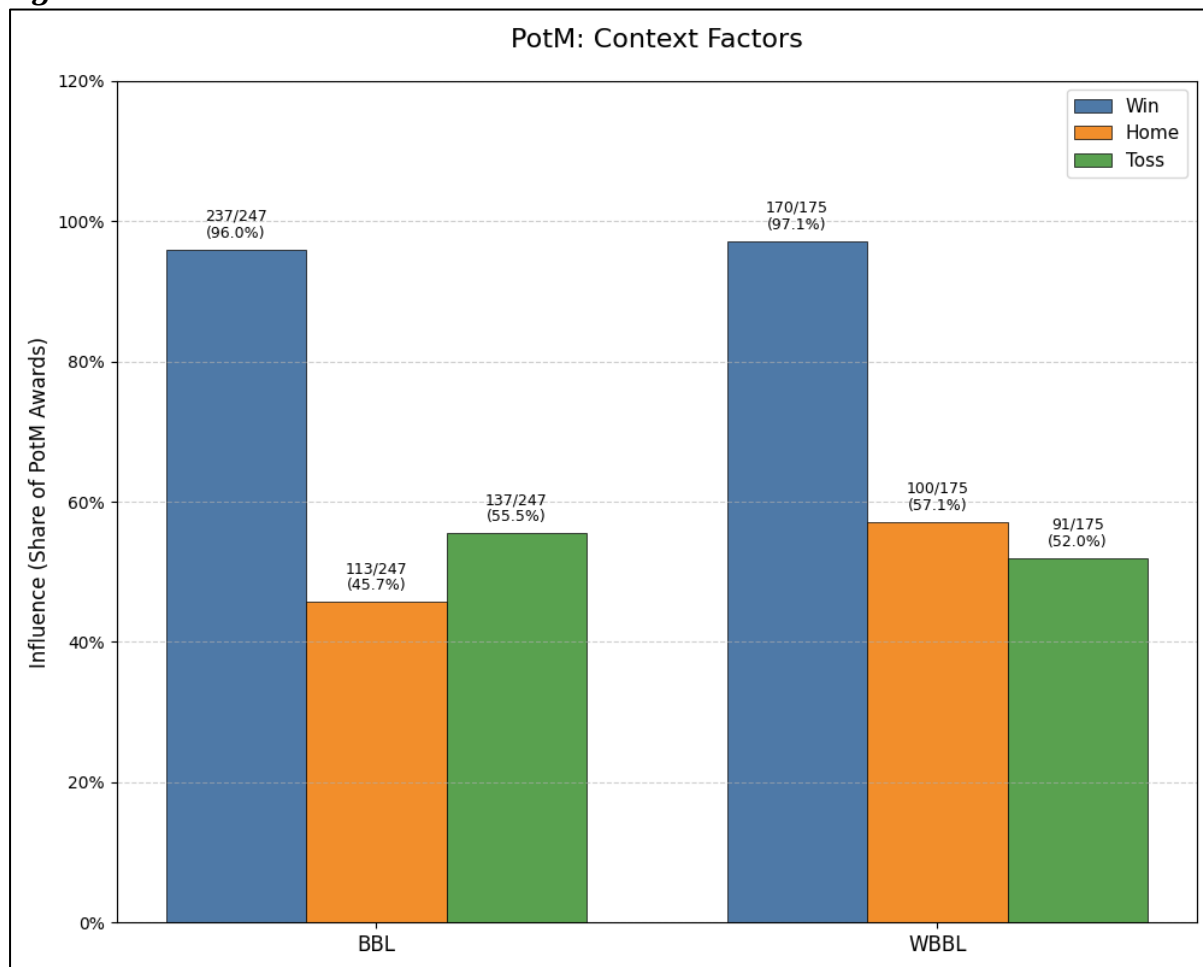
In the data, PotM's scored a median of 51 runs in a match, compared to a median of 5 for non-winners. 72.99% of recipients recorded at least 20% of their teams total run share, and 19.91% of recipients had less then 1% of the team share. Unsurprisingly, runs and run

share are highly correlated ($r \sim 0.90$), this feature displays player influences proportional to team performance. High strike rates (runs/balls faced) also bolster batting impact.

Figure 2: Bowling contribution:



Generally, the more wickets a player took, the higher their probability of PotM choice, observing an exponential relationship. It was also found that not all wickets are equal, bowler economy provides insight for average bowls/wicket. This rate influences PotM, has less effect than wicket count (*appendix 1*).

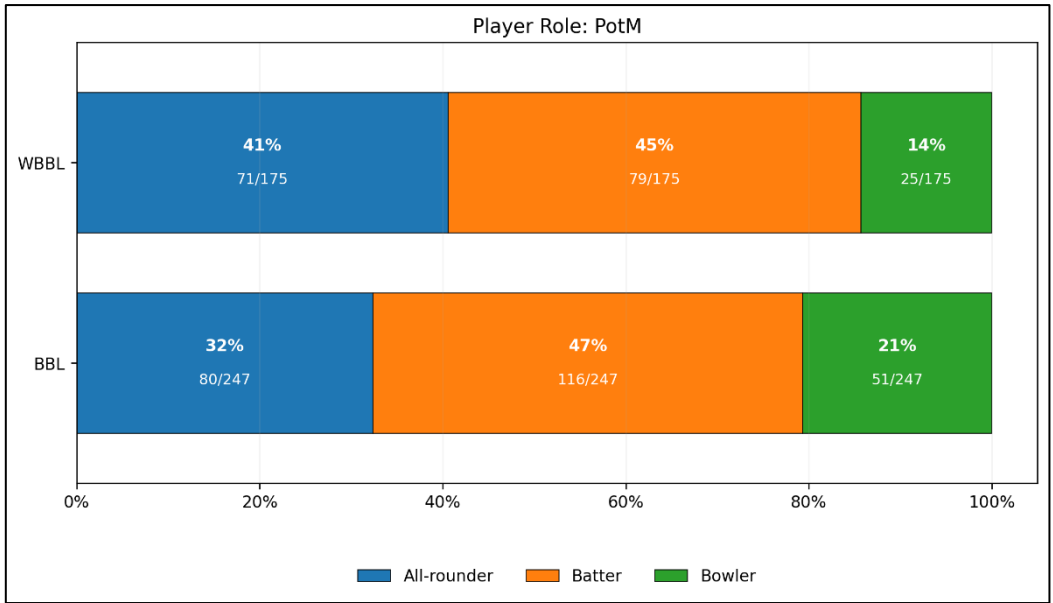
Match Information:**Figure 3: Contextual Factors:**

The strongest context factor affecting PotM was whether the player's team won the match.

$\frac{407}{422}$, ~ 96.45% of all of all PotM winners were from the winning team. This recognised that any model will heavily favour players on the match-winning team, and will struggle to recognise PotM impact from losing teams. Contrasting, home team proportion is

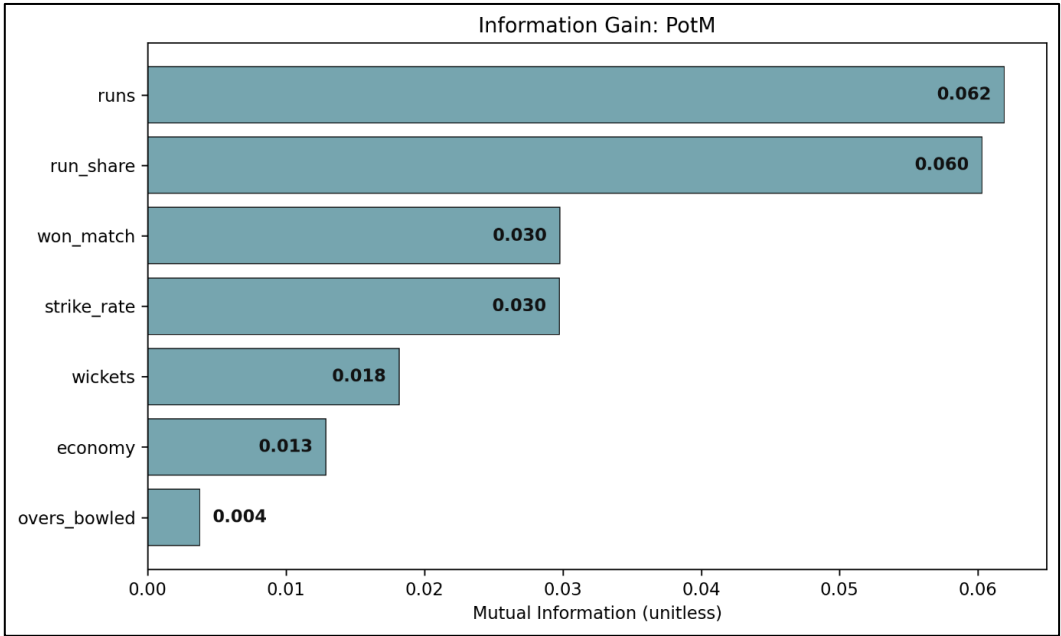
$\frac{213}{424}$ ~50.23%, toss-winner and PotM was a similar result.

Figure 4: Player Role Contribution



Looking at PotM distributions across player roles, we observe batters having a 46.21%, all rounders 35.78% and bowlers having 18.01% share of the PotM awards. Player role influence reinforces that multi-dimensional contributions are highly valued in PotM selection.

Figure 5: Information Gain



To quantify which features carry the most predictive signal for PotM, mutual information (MI) was calculated between each feature and PotM outcome. MI is a measure of how much a feature reduces uncertainty about the another (PotM). Runs scored and run share emerged as the two top predictors by wide margin. Other informative features were the

Won Match flag, strike rate and to a lesser effect wicket taken. We additionally measured correlation between features, which can lead to inflated variance for feature coefficients (*appendix 2*).

2.3- Feature Engineering

The insights from the exploratory analysis guided the creation of several engineered features to improve model learning. In total, the final modelling dataset includes 32 features, including those engineered from raw data. Notable engineered features include: ***Full Engineered Feature List in appendix [3]***

Section 3: Modelling Methodology Justification

In addressing a discrete choice task, four complementary modelling approaches were identified. Each possessing characteristics approach the problem. Given the nature of the data, potential non-linear effects, high dimensionality and correlated features. The models are:

- Penalised Logistic Regression (LR)
- Conditional Generalized Linear Model (GLM)
- Generalized Additive Model (GAM)
- Random Forest (RF) ensemble.

3.1 – Penalised Logistic Regression:

LR provides a natural starting point for binary choice prediction. It estimates the probability of a player winning PotM as the logistic function of features, coefficients offering an interpretable measure of each feature's effect (*ceteris paribus*). Given the high-dimensional space, with measured feature correlation, and identified non-linearities. Simple logistic regression hinders, using it would lead to overfitting, unreliable coefficients, and extensive multicollinearity.

We use a penalisation term to minimise the loss function of the logistic regression. Methods: L1 (lasso), L2 (ridge), and a combination of ridge and lasso (elastic net) penalisation were compared to address this.

L1,L2 penalty:

$$L1 = \lambda \sum_{i=1}^p |\beta_i| \quad , \quad L2 = \lambda \sum_{i=1}^p \beta_i^2$$

Elastic

Net

Penalty:

$$EN = \lambda \left(\alpha \sum_{i=1}^p |\beta_i| + \frac{1-\alpha}{2} \sum_{i=1}^p \beta_i^2 \right)$$

LR Features:

$$\eta_i = \beta_0 + x_i^\top \beta_i, \quad p_i = \sigma(\eta_i) \frac{1}{1+e^{-\eta_i}}$$

Model Function:

$$\min_{\beta_0, \beta_k} \frac{1}{n} \sum_{i=1}^n [-y_i \log(p_i) - (1-y_i) \log(1-p_i)] + \lambda \left[\frac{1-\alpha}{2} \|\beta\|_2^2 + \alpha \|\beta\|_1 \right]$$

For Net elastic penalisation lambda controls the overall strength of the penalty whilst alpha represents the mixing weighting parameter balancing L1 and L2. Naturally $0 \leq \alpha \leq 1$.

3.2 – Conditional Generalised Linear Model (GLM)

An important aspect of the problem is the grouped structure with one winning outcome. Standard logistic regression could predict more than one player per game with >50% probability, which breaks the 1 outcome requirement. Incorporating per-match competition we use a conditional structure ensuring match probabilities sum to 1. This is analogous to discrete choice or link formulation, with:

Logit link,

$$P_{im} = \frac{e^{(\eta_{im})}}{\sum_{j \in 1} e^{(\eta_{im})}}$$

Probit link,

$$P_{im} = \Phi(\eta_{im}), \text{ where } \Phi \text{ is the standard normal CDF}$$

Cauchit link,

$$P_{im} = \text{CDF}_{\text{Cauchy}}(\eta_{im}) = \frac{1}{\pi} \tan^{-1}(\eta_{im}) + 0.5$$

Loglog link,

$$P_{im} = 1 - e^{-e^{\eta_{im}}}$$

Cloglog link,

$$P_{im} = 1 - e^{-e^{-\eta_{im}}}$$

Whilst logit is standard for multinomial choice, capturing unobserved factors (probit) or extreme outliers (cauchit) can offer improved insight. Therefore a comprehensive approach is used.

3.3 – Generalised Additive Model (GAM)

The exploratory data analysis revealed non-linear relationships. Generalized Additive Models (GAMs) provide a flexible extension of logistic regression by allowing non-linear functions of each predictor. For a GAM, the log-odds of PotM are defined as:

$$\log\left(\frac{P(y = 1|X)}{1 - P(Y = 1|X)}\right) = \beta_o + f_1(x_1) + f_2(x_2) + \dots f_p(x_p)$$

Where each $f_p(x_p)$ is a smooth function fit to the data. This means the model can learn non-linear relationships. Smoothing splines are key for reliable predictors. Further applying penalty for the smooth terms avoid overfitting complex relationships. It is expected that GAM's excel in calibrating probabilities, leveraging its capture of nuanced data.

3.4 – Random Forest (RF) ensemble

Finally, a RF model, identifying non-parametric relationships capture complex interactions that simple regression methods fail. Bootstrapping the tree learning process aggregate predictions to inform an orbicular perspective.

Strengths:

- Handles complex non-linear relationships and feature interaction. Dynamic outlier handling, and dimensionality assessment.

Weakness:

- Poor interpretability. It doesn't provide simple coefficients, though general feature importance can be estimated.

The RF model is used to set a performance benchmark for the other models.

Summary

- M1: Provides a simple, transparent baseline model that assumes linear relationships and uses penalty feature selection.
- M2: Specialised model that removed cross-match bias by comparing players within the same match.
- M3: A flexible interpretable model that captures non-linear effects, whilst maintaining interpretability.
- M4: A powerful approach used as a performance benchmark to capture complex relationships.

The main goal is to compare these diverse models to understand the trade-off between simplicity and accuracy. Observing if complex approaches outperform simpler linear approaches, determining the most effective and appropriate approach.

Section 4: Model Findings and Comparative Analysis

After training the models (using cross-validation and hold-out techniques) to tune hyperparameters), each model's performance is evaluated. Given the extreme class imbalance (1 positive case of 22 players), overall accuracy would be misleading, as simply predicting no for all players would yield a 95.5% accuracy.

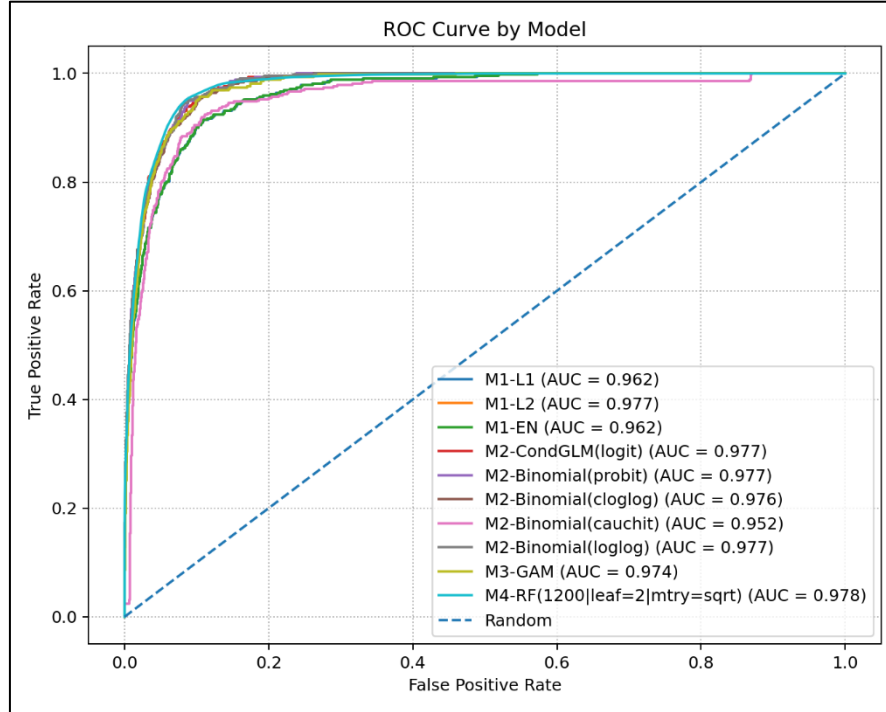
- Ranking metrics:
 - Mean Reciprocal Rank (MRR) is the average of the reciprocal of the rank of the actual winner. Where $MRR=1$ represents the winner being correctly identified.
 - NDCG@3 looks at the top 3 rankings and gives a weighted score $0 \leq NDCG \leq 1$, the closer to 1 the higher the credibility of specification.
- Probability Metrics:
 - Log loss (cross entropy) penalises the model for assigning low probability to the true winner, Los-loss=0 represents the best probability estimate.
 - Brier score is the ε_{μ}^2 of probability predictions, BS = 0 represent predicted probabilities specifying actual binary outcome flags.
 - Expected Calibration Error (ECE) measures the difference between predicted probabilities and actual frequencies using decile binning approach. A low ECE means that probability estimates are practically applicable.

For this specific problem, where one winner is chosen from 22 candidates the standard classification metric Recall (or sensitivity or True Positive Rate) is mathematically equivalent to Top-1 Accuracy [**appendix 4**] as both measure the proportion of times the single true winner was correctly identified. Additionally, I considered the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) to measure each model's ability to discriminate points between PotM winners and non-winners. The complete performance data is available in [**appendix 5-6**].

For fair comparison, all models are evaluated on the same held-out data (via cross-validation and out-of-fold methods). Additionally, we analyse model performance under three different feature scaling methods to test the sensitive to data pre-processing.

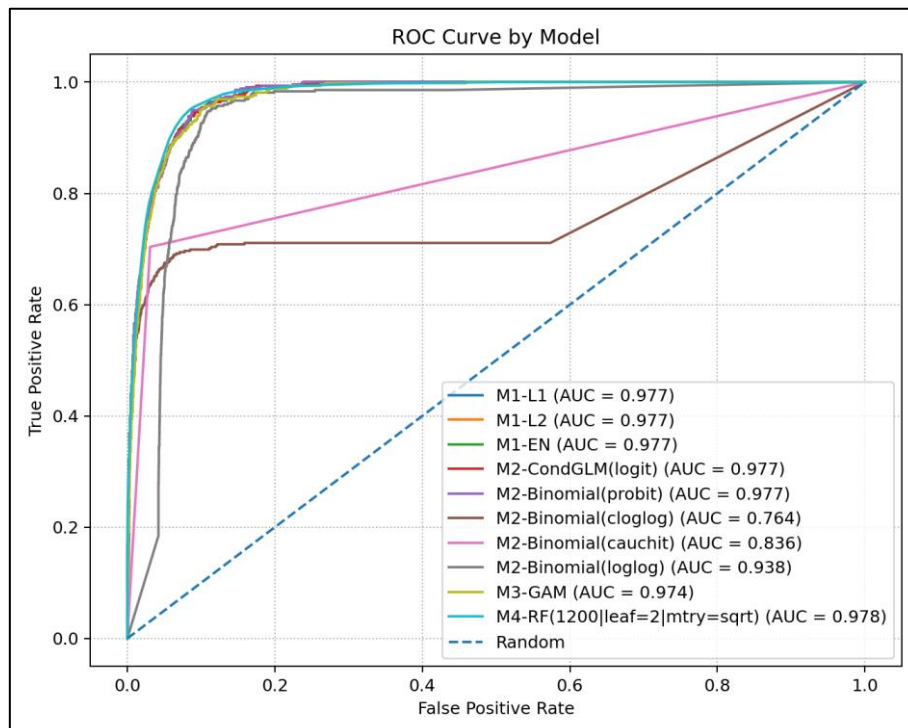
- No scaling (raw features):
 - Using the original units of each feature. More suitable for ensemble methods like Random Forests, lack scaling for linear models.

Figure 6: ROC curve for Raw Features



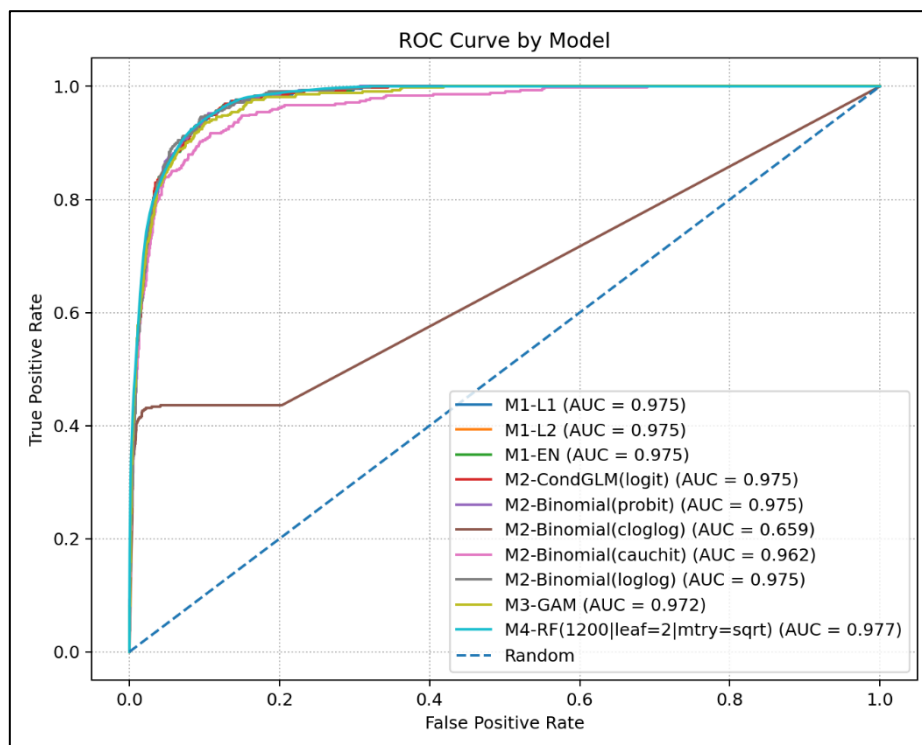
- Global Scaling:
 - Using a standard scalar on features across the entire dataset $\left(\frac{x_i - \mu_i}{\sigma_i}\right)$.

Figure 7: ROC curve for Global Scaling



- Match level Normalisation:
 - Standardising features within each match. Providing an effective measure of player performance for in game contexts.

Figure 8: ROC curve for Match Z-scores



Full Results in appendix X.

It is observed that feature scaling had a notable influence on logistic models, performing best with global standardization, whereas the per-match z-scoring while conceptually appealing, scaling for the match without taking into consideration how good that game was historically. The conditional GLM model inherently works within-match so it was less affected by scaling. Random Forest saw its best performance on the raw data.

Model 1: Penalised Logistic Regression Results

Scenario	Model	Top-1	Top-3	MRR	NDCG3	Logloss	Brier	ECE
Global Scale	EN	0.6896	0.9360	0.8176	0.8376	0.8853	0.0203	0.0070
Raw Data	L2	0.6706	0.9384	0.8076	0.8312	0.8832	0.0202	0.0072

The elastic net (EN) penalty on globally scaled data produced the best ranking performance, correct identifying ~69% of matches. This model also demonstrated excellent discriminatory power with an AUC of 0.9769. However, its probability estimates were overstated as shown with a calibration slope of 1.36.

Model 2: Conditional GLM results

Scenario	Model	Top-1	Top-3	MRR	NDCG3	Logloss	Brier	ECE
Global Scale	Logit	0.6896	0.9336	0.8175	0.8364	0.8856	0.0203	0.0070
Raw Data	Probit	0.6682	0.9431	0.8055	0.8314	0.8644	0.0202	0.0043
Match Z-score	Cauchit	0.6209	0.8981	0.7664	0.7868	1.2505	0.0240	0.0155
Raw Data	Loglog	0.6825	0.9455	0.8133	0.8382	0.8735	0.0204	0.0027

Unsurprisingly the logit link for conditional GLM mirrors L2 logistic regression. This is because both models apply the same transformation process.

$$LR = \sigma(\eta_i) \frac{1}{1 + e^{-\eta_i}} \quad , \quad \text{Logit}_{im} = \frac{e^{(\eta_{im})}}{\sum_{j \in 1} e^{(\eta_{im})}}$$

The probit link, assumed a normal distribution of unobserved factors, offered slight improvement in probability estimation with reduced log loss and ECE. Surprisingly the asymmetric loglog link on raw data achieved the best calibration and strong shortlisting (Top-3 = 94.55%) performance (AUC matching L2 LR), making it a reliable choice for generating probabilities.

Model 3: GAM Results

Scenario	Model	Top-1	Top-3	MRR	NDCG3	Logloss	Brier	ECE
Raw Data / Global Scale	GAM	0.6730	0.9502	0.8093	0.8380	0.9027	0.0211	0.0024
Match Z-score	GAM	0.6659	0.9123	0.7955	0.8121	0.9846	0.0215	0.0024

The GAM delivers the strongest overall performance under raw data and global scaling inputs. Achieving a balance between exceptional performance (top3 0.9502) and accurate probability calibration (ECE 0.0024).

Under z-scored inputs, it maintains strong rankings. Translating the dynamic versatility of GAM's, capturing accurate rankings and producing faithful probabilities.

Model 4: RF Results

Scenario	Model	Top-1	Top-3	MRR	NDCG3	Logloss	Brier	ECE
Raw Data	RF	0.6777	0.9502	0.8108	0.8394	0.8360	0.0195	0.0045
Global Scale	RF	0.6754	0.9502	0.8093	0.8382	0.8367	0.0195	0.0045
Match Z-score	RF	0.6564	0.9218	0.7935	0.8149	0.8955	0.0204	0.0044

The RF model excelled on raw data, achieving the lowest log loss (0.8360) and highest AUC (0.978) of any model, indicating excellent discriminatory power. Ranking metrics were top-tier with an MRR of 0.8108 implying an average true PotM rank of ~ 1.23 and a top-3 matching the GAM at 95%. While its calibration was strong (ECE = 0.0045) it was slightly edged out by the GAM and loglog GLM, showcasing a minor trade-off between raw predictive power and precision of its probability estimation.

Section 5: Discussion and Implications

This section interprets the modelling results in practical and theoretical terms. Focusing on key findings, implications for discrete choice prediction and evaluation of PotM outcomes. It also reflects on the strengths and weaknesses of prediction methodology, identifying model strengths and limitations.

5.1 – Key findings

Performance is quantifiable, the results infer that it is feasible to predict Player of the Match (PotM) outcomes using objective performance data. The evaluation framework, finds equilibrium between ranking quality and probability calibration identifies model types and scaling strategies.

GAM (model 3) emerged as the most balanced and replicable model, achieving strong ranking metrics.

- Strongest top-3 accuracy, where PotM is found in the top-3 95.02% of the time.
- MRR = 0.8093: where $\frac{1}{0.8093}$, averaging correct PotM rank 1.24
- Calibration ECE = 0.0024 throughout all data processing structures.

Conditional GLM (model 2) shows flexible behaviour across link function.

- The loglog and probit links particularly complement the findings of model 3. Achieving comparable ranking metrics with the added benefit of reduced log loss.

Logistic regression (model 1) performed as a competitive baseline, when properly scaled.

- The elastic net variant achieved the highest top 1 accuracy (68.96%) using globally scaled data. Confirming that regularisation improves ranking under linear constraint.
- Probability calibration was uncompetitive with GAM and GLM models.
- Model 1 log loss was similar to improvements found in GLM's.

Random forest (model 4) achieved the highest ranking and probability performance on raw inputs.

- Highest AUC = 0.9780.
- Top 3 = 95.02%
- MRR = 0.8108, average rank 1.23 for true PotM.
- Lowest logloss (0.8360). Reinforcing the assumption RF would provide a strong benchmark.

5.2 – Practical Implications

The findings carry several important implications for applying machine learning modelling to PotM or similar discrete choice settings:

1. Models need to be match aware.
 - a. Methods like conditional GLM's naturally handle the one-winner-per structure. Models that ignore this risk exceed the sample space.
2. Context Relevance.
 - a. Per match z-scoring, while logically sound in theory, reduced accuracy. By removing the context of a player's performance relative to historical data, this over-normalisation made it harder for models to identify definitive signal.
3. Data Processing.
 - a. Scaling must be matched to the model family. Linear models benefited most from global standardisation, while the tree-based RF ensemble worked best on raw input.
4. The "best" model:
 - a. For reliable probabilities (bookmaker odds) the GAM or a GLM with probit/loglog link is most appropriate.
 - b. For accurate ranking and short listing (broadcast commentary) the Random Forest of raw inputs delivers the best performance.

5.3 – Assumptions and limitations

- **Simplified representation:** The models rely solely on summary data and do not capture the broader narrative. Heterogenous variable incorporation such as weather, clutch moments, or crowd size potentially add value.
- **Team-win feature:** For in game broadcasting the inclusion of the match result as a feature introduces explicit bias, limiting the model's features to account for this is better suited for live forecasting without a supplementary win prediction model.
- **Data Grouping:** The analysis assumes the criteria for awarding PotM has remained stable across all seasons and applies equally to BBL and WBBL matches.

Section 6: Conclusion and Recommendations

This project demonstrates that the BBL Player of the Match award can be predicted with accuracy and reliability using only objective performance statistics. Penalised logistic regression provided the best correct-ranking accuracy, while generalised additive and Random Forest models delivered improved calibration and consistent top-three rankings. The analysis confirms that batting contribution and team victory are substantial determinants of T20 PotM, with bowling excellence and all-round performance strong supporters.

From a broader perspective, the report illustrates how statistical learning techniques can increase transparency in award systems, complementing human judgement and boost fan engagement. Future work should incorporate temporal context, explore other flexible models (gradient-boosted trees), and test the framework on other leagues or sports. A simple “PotM impact score” derived from the model's coefficients could provide a user-friendly metric for commentators, coaches and fans.

Overall, the findings support the adoption of data-driven methods in recognising sporting excellence and suggest that objective metrics can meaningfully inform awards traditionally seen as subjective.

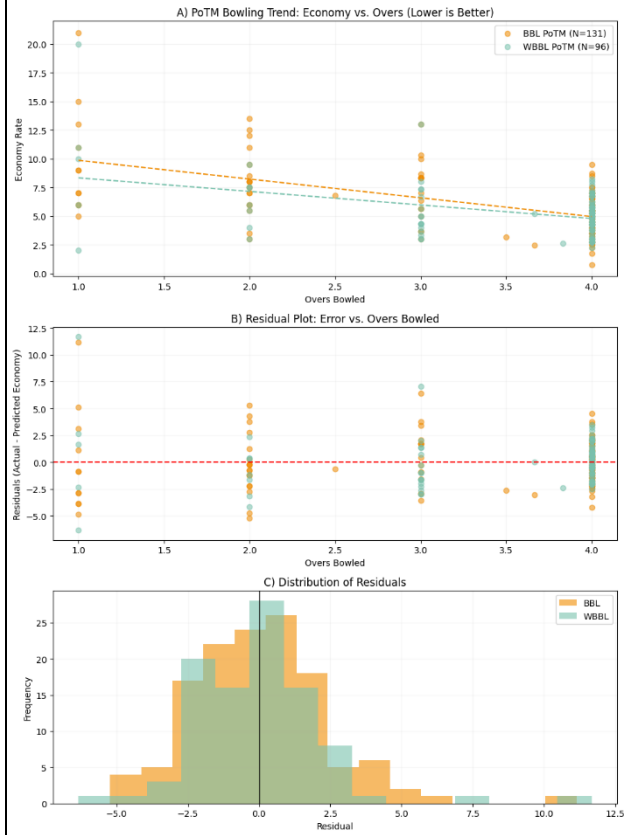
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Appendix

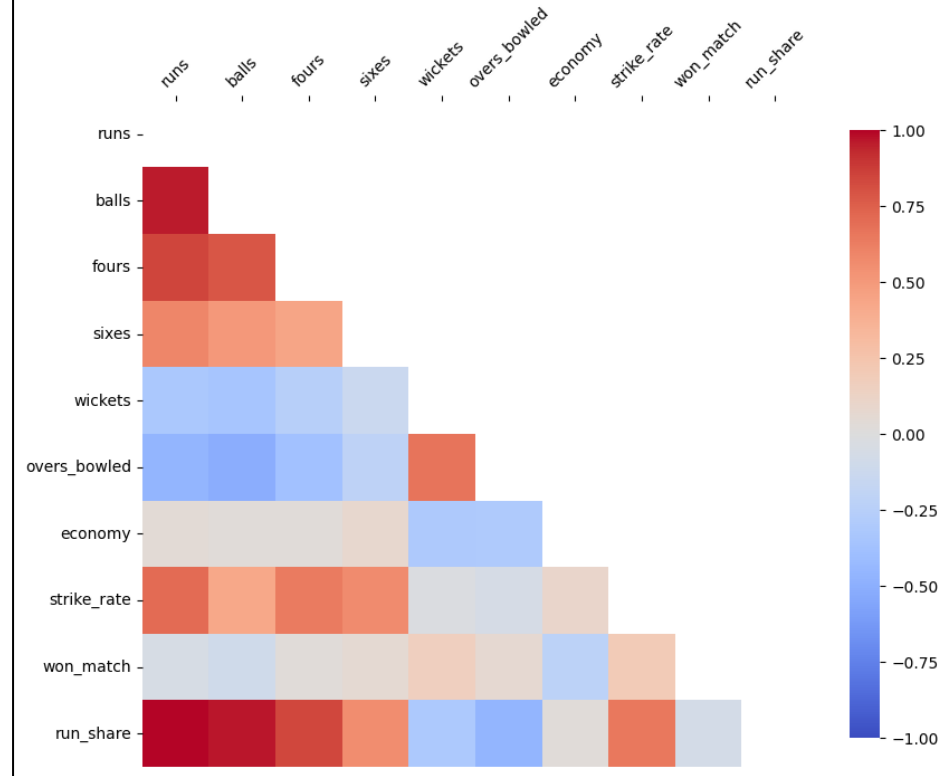
[1]

Bowling PoTM Context: OLS Trend & Residual Analysis



[2]

Spearman Correlation — Performance Features



[3]

Feature Name	Description	Process
Game_ID	A unique identifier for each match.	Generated by combining league, season, and match number.
POTM	(Target Variable) A binary flag (1/0) indicating if the player won the PotM award.	Derived by comparing player name to the official PotM name.
Team_Won	A binary flag (1/0) indicating if the player's team won the match.	Derived from the official match winner record.
Team_Home	A binary flag (1/0) indicating if the player's team was the designated home team.	Derived from the official home team record.
Batter, Bowler, All_Rounder	Binary flags indicating the player's role in the match.	Determined by whether the player batted, bowled, or did both.

[4]

scenario	model	top1	tpr	recall	sensitivity
scaled	M1-EN	0.6896	0.6896	0.6896	0.6896
scaled	M2-CondGLM(logit)	0.6896	0.6896	0.6896	0.6896
scaled	M1-L2	0.6896	0.6896	0.6896	0.6896
raw	M2-Binomial(loglog)	0.6825	0.6825	0.6825	0.6825
scaled	M1-L1	0.6825	0.6825	0.6825	0.6825
raw	M3-GAM	0.6730	0.6730	0.6730	0.6730
scaled	M3-GAM	0.6730	0.6730	0.6730	0.6730
raw	M4-RF(1200 leaf=2 mtry=sqrt)	0.6777	0.6706	0.6706	0.6706
raw	M1-L2	0.6706	0.6706	0.6706	0.6706
raw	M2-CondGLM(logit)	0.6706	0.6706	0.6706	0.6706
scaled	M4-RF(1200 leaf=2 mtry=sqrt)	0.6754	0.6706	0.6706	0.6706
raw	M2-Binomial(probit)	0.6682	0.6682	0.6682	0.6682
scaled	M2-Binomial(probit)	0.6682	0.6682	0.6682	0.6682
zscore	M3-GAM	0.6659	0.6659	0.6659	0.6659
raw	M2-Binomial(cloglog)	0.6588	0.6588	0.6588	0.6588
zscore	M1-L1	0.6588	0.6588	0.6588	0.6588
raw	M1-EN	0.6564	0.6564	0.6564	0.6564
raw	M1-L1	0.6564	0.6564	0.6564	0.6564
zscore	M2-CondGLM(logit)	0.6564	0.6564	0.6564	0.6564
zscore	M1-L2	0.6564	0.6564	0.6564	0.6564

Run_Share	Player's percentage of team runs scored.	Calculated as Runs_Scored / Team_Total_Runs.	zscore	M4-RF(1200 leaf=2 mtry=sqrt)	0.6564	0.6540	0.6540	0.6540
Wicket_Impact	Exponential weighting of wicket counts.	A transformation to capture the non-linear trend observed in the EDA.	zscore	M1-EN	0.6540	0.6540	0.6540	0.6540
			zscore	M2-Binomial(probit)	0.6517	0.6517	0.6517	0.6517
			zscore	M2-Binomial(loglog)	0.6517	0.6517	0.6517	0.6517
			zscore	M2-Binomial(cauchit)	0.6209	0.6209	0.6209	0.6209
			raw	M2-Binomial(cauchit)	0.6114	0.6114	0.6114	0.6114
			scaled	M2-Binomial(loglog)	0.5758	0.5758	0.5758	0.5758
			scaled	M2-Binomial(cloglog)	0.5569	0.5569	0.5569	0.5569
			scaled	M2-Binomial(cauchit)	0.5237	0.5355	0.5355	0.5355
			zscore	M2-Binomial(cloglog)	0.4147	0.4100	0.4100	0.4100

[5]

scenario	model	top1	top3	mrr	ndcg3	Group logloss	brier	ece	Cal slope	Cal intercept	Entropy mean	mass dev_ind
raw	L1	0.6564	0.9005	0.7856	0.8011	1.1054	0.023	0.0098	1.3456	0.497	1.4757	0.398
scaled	L1	0.6825	0.9336	0.8142	0.8341	0.8846	0.0202	0.0074	1.3583	0.3776	1.1683	0.3319
zscore	L1	0.6588	0.9336	0.7964	0.8223	0.9561	0.0217	0.005	1.2164	0.2259	1.151	0.2643
raw	L2	0.6706	0.9384	0.8076	0.8312	0.8832	0.0202	0.0072	1.3711	0.3937	1.1787	0.3325
scaled	L2	0.6896	0.9336	0.8175	0.8364	0.8856	0.0203	0.007	1.3624	0.3825	1.1713	0.3333
zscore	L2	0.6564	0.936	0.7955	0.8226	0.9577	0.0218	0.0048	1.2125	0.2225	1.1521	0.2651
raw	EN	0.6564	0.9005	0.7848	0.8005	1.1049	0.023	0.01	1.3444	0.4948	1.474	0.3976
scaled	EN	0.6896	0.936	0.8176	0.8376	0.8853	0.0203	0.007	1.3615	0.3815	1.171	0.3327
zscore	EN	0.654	0.936	0.7946	0.822	0.9565	0.0218	0.0048	1.2152	0.2249	1.1519	0.2647
raw	logit	0.6706	0.9384	0.8076	0.8312	0.8832	0.0202	0.0072	1.3711	0.3937	1.1787	0.3325
scaled	logit	0.6896	0.9336	0.8175	0.8364	0.8856	0.0203	0.007	1.3624	0.3825	1.1713	0.3333
zscore	logit	0.6564	0.936	0.7955	0.8226	0.9577	0.0218	0.0048	1.2125	0.2225	1.1521	0.2651
raw	probit	0.6682	0.9431	0.8055	0.8314	0.8644	0.0202	0.0043	1.299	0.2736	1.0507	0.3269
scaled	probit	0.6682	0.9431	0.8055	0.8314	0.8644	0.0202	0.0043	1.299	0.2736	1.0507	0.3269
zscore	probit	0.6517	0.9408	0.7952	0.8244	0.9536	0.022	0.0034	1.1515	0.1406	1.0588	0.2522
raw	cauchit	0.6114	0.8744	0.7526	0.7665	1.3448	0.0256	0.0163	1.3233	0.6211	1.8158	0.588
scaled	cauchit	0.5237	0.737	0.6498	0.6514	3.6033	0.0276	0.0126	0.1656	-1.5783	0.8041	0.6588
zscore	cauchit	0.6209	0.8981	0.7664	0.7868	1.2505	0.024	0.0155	1.323	0.6264	1.7691	0.4234
raw	loglog	0.6825	0.9455	0.8133	0.8382	0.8735	0.0204	0.0027	1.2369	0.1878	0.9756	0.3115
scaled	loglog	0.5758	0.8578	0.7299	0.741	1.3084	0.024	0.0042	0.8545	-0.1531	1.1088	1.0987
zscore	loglog	0.6517	0.936	0.7954	0.8224	0.9767	0.0222	0.0028	1.0869	0.0716	0.9996	0.2394
raw	cloglog	0.6588	0.9313	0.7987	0.8214	0.9294	0.0207	0.0107	1.4216	0.514	1.2917	0.3379
scaled	cloglog	0.5569	0.7393	0.6673	0.6636	2.1148	0.0263	0.0046	0.4398	-1.3195	1.5105	0.4497
zscore	cloglog	0.4147	0.5355	0.5133	0.486	3.0007	0.0322	0.0059	0.1991	-2.2257	1.8021	0.5746
raw	GAM	0.673	0.9502	0.8093	0.838	0.9027	0.0211	0.0024	1.0962	0.0892	0.9818	0.3703

scaled	GAM	0.673	0.9502	0.8093	0.838	0.9027	0.0211	0.0024	1.0962	0.0892	0.9818	0.3703
zscore	GAM	0.6659	0.9123	0.7955	0.8121	0.9846	0.0215	0.0024	0.9266	-0.0691	0.8833	0.3077
raw	RF	0.6777	0.9502	0.8108	0.8394	0.836	0.0195	0.0045	1.3381	0.2848	1.0268	0.3166
scaled	RF	0.6754	0.9502	0.8093	0.8382	0.8367	0.0195	0.0045	1.3429	0.2884	1.029	0.315
zscore	RF	0.6564	0.9218	0.7935	0.8149	0.8955	0.0204	0.0044	1.2335	0.2205	1.0493	0.2582

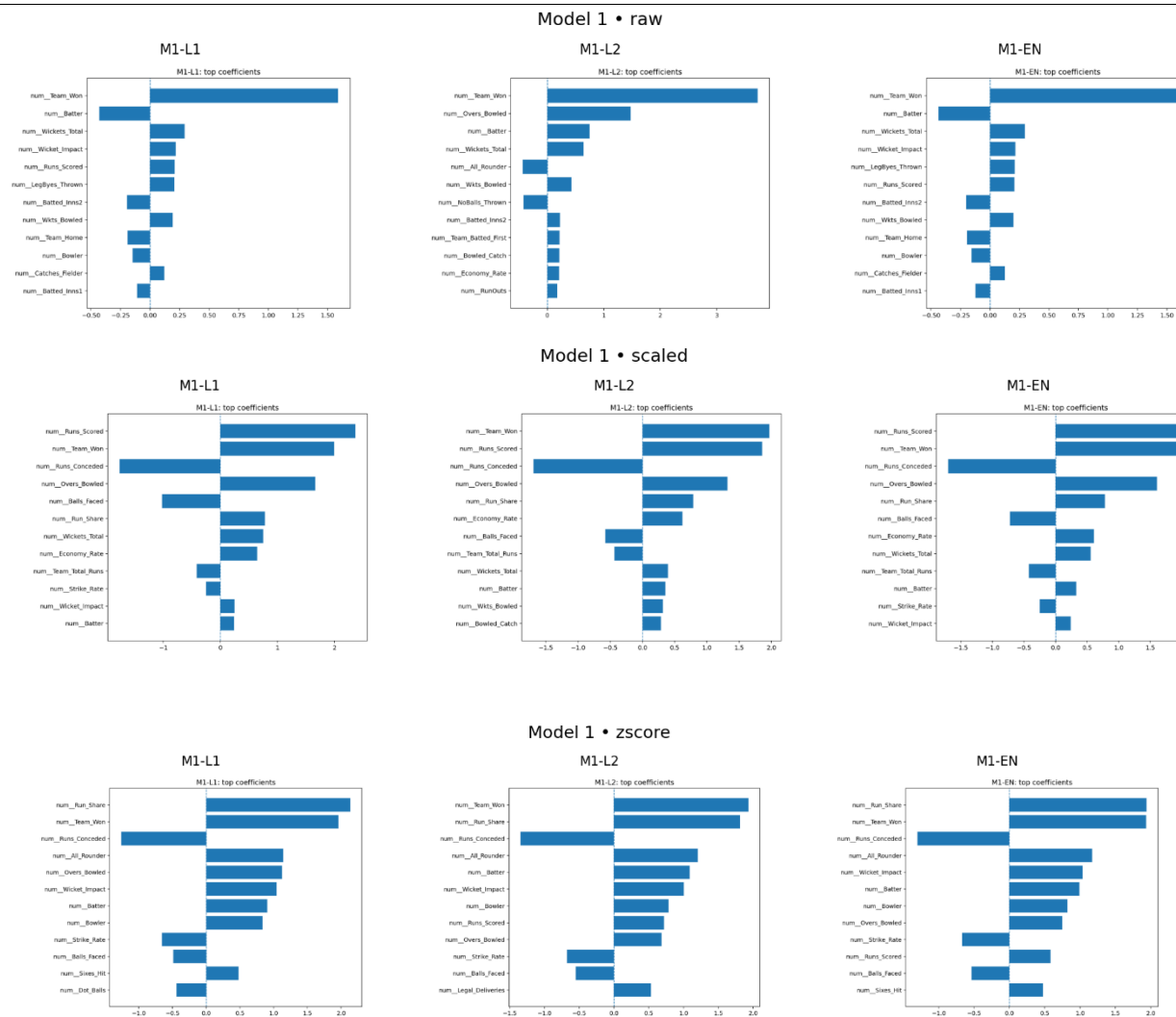
[6]

scenario	model	Auc roc	tp	fn	fp	tn	tpr	recall	Sens itivity	fnr	fpr	tnr	prevalence	accuracy	f1
raw	L1	0.9618	277	145	145	8493	0.6564	0.6564	0.6564	0.3436	0.0168	0.9832	0.0466	0.968	0.6564
scaled	L1	0.9769	288	134	134	8504	0.6825	0.6825	0.6825	0.3175	0.0155	0.9845	0.0466	0.9704	0.6825
zscore	L1	0.9746	278	144	144	8494	0.6588	0.6588	0.6588	0.3412	0.0167	0.9833	0.0466	0.9682	0.6588
raw	L2	0.977	283	139	139	8499	0.6706	0.6706	0.6706	0.3294	0.0161	0.9839	0.0466	0.9693	0.6706
scaled	L2	0.9768	291	131	131	8507	0.6896	0.6896	0.6896	0.3104	0.0152	0.9848	0.0466	0.9711	0.6896
zscore	L2	0.9745	277	145	145	8493	0.6564	0.6564	0.6564	0.3436	0.0168	0.9832	0.0466	0.968	0.6564
raw	EN	0.9618	277	145	145	8493	0.6564	0.6564	0.6564	0.3436	0.0168	0.9832	0.0466	0.968	0.6564
scaled	EN	0.9769	291	131	131	8507	0.6896	0.6896	0.6896	0.3104	0.0152	0.9848	0.0466	0.9711	0.6896
zscore	EN	0.9746	276	146	146	8492	0.654	0.654	0.654	0.346	0.0169	0.9831	0.0466	0.9678	0.654
raw	logit	0.977	283	139	139	8499	0.6706	0.6706	0.6706	0.3294	0.0161	0.9839	0.0466	0.9693	0.6706
scaled	logit	0.9768	291	131	131	8507	0.6896	0.6896	0.6896	0.3104	0.0152	0.9848	0.0466	0.9711	0.6896
zscore	logit	0.9745	277	145	145	8493	0.6564	0.6564	0.6564	0.3436	0.0168	0.9832	0.0466	0.968	0.6564
raw	probit	0.9771	282	140	140	8498	0.6682	0.6682	0.6682	0.3318	0.0162	0.9838	0.0466	0.9691	0.6682
scaled	probit	0.9771	282	140	140	8498	0.6682	0.6682	0.6682	0.3318	0.0162	0.9838	0.0466	0.9691	0.6682
zscore	probit	0.9747	275	147	147	8491	0.6517	0.6517	0.6517	0.3483	0.017	0.983	0.0466	0.9675	0.6517
raw	cauchit	0.9517	258	164	164	8474	0.6114	0.6114	0.6114	0.3886	0.019	0.981	0.0466	0.9638	0.6114
scaled	cauchit	0.8364	226	196	196	8442	0.5355	0.5355	0.5355	0.4645	0.0227	0.9773	0.0466	0.9567	0.5355
zscore	cauchit	0.9621	262	160	160	8478	0.6209	0.6209	0.6209	0.3791	0.0185	0.9815	0.0466	0.9647	0.6209

raw	loglog	0.9774	288	134	134	8504	0.6825	0.6825	0.6825	0.3175	0.0155	0.9845	0.0466	0.9704	0.6825
scaled	loglog	0.9377	243	179	179	8459	0.5758	0.5758	0.5758	0.4242	0.0207	0.9793	0.0466	0.9605	0.5758
zscore	loglog	0.9751	275	147	147	8491	0.6517	0.6517	0.6517	0.3483	0.017	0.983	0.0466	0.9675	0.6517
raw	cloglog	0.9758	278	144	144	8494	0.6588	0.6588	0.6588	0.3412	0.0167	0.9833	0.0466	0.9682	0.6588
scaled	cloglog	0.7638	235	187	187	8451	0.5569	0.5569	0.5569	0.4431	0.0216	0.9784	0.0466	0.9587	0.5569
zscore	cloglog	0.6588	173	249	249	8389	0.41	0.41	0.41	0.59	0.0288	0.9712	0.0466	0.945	0.41
raw	GAM	0.974	284	138	138	8500	0.673	0.673	0.673	0.327	0.016	0.984	0.0466	0.9695	0.673
scaled	GAM	0.974	284	138	138	8500	0.673	0.673	0.673	0.327	0.016	0.984	0.0466	0.9695	0.673
zscore	GAM	0.9721	281	141	141	8497	0.6659	0.6659	0.6659	0.3341	0.0163	0.9837	0.0466	0.9689	0.6659
raw	RF	0.978	283	139	139	8499	0.6706	0.6706	0.6706	0.3294	0.0161	0.9839	0.0466	0.9693	0.6706
scaled	RF	0.9779	283	139	139	8499	0.6706	0.6706	0.6706	0.3294	0.0161	0.9839	0.0466	0.9693	0.6706
zscore	RF	0.9766	276	146	146	8492	0.654	0.654	0.654	0.346	0.0169	0.9831	0.0466	0.9678	0.654

[7]

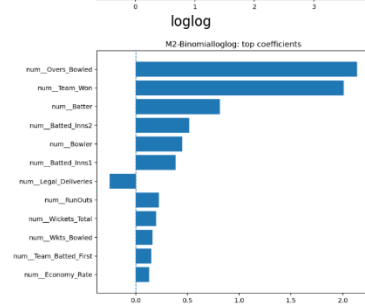
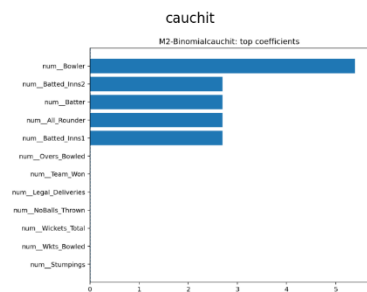
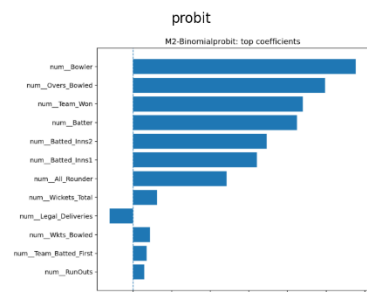
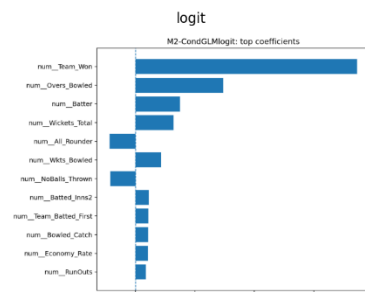
M1 – Coeff



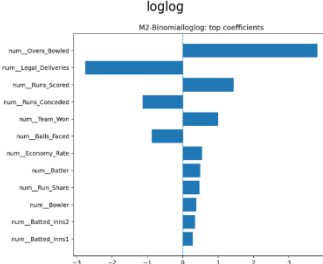
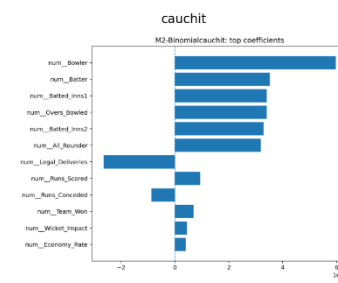
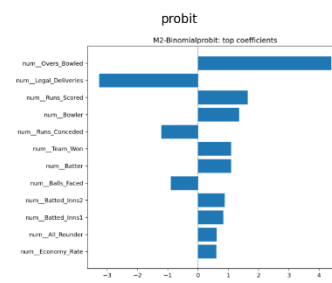
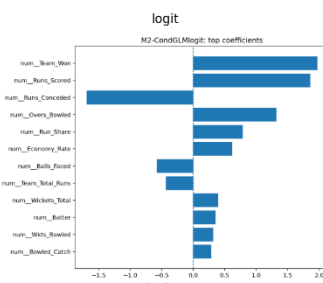
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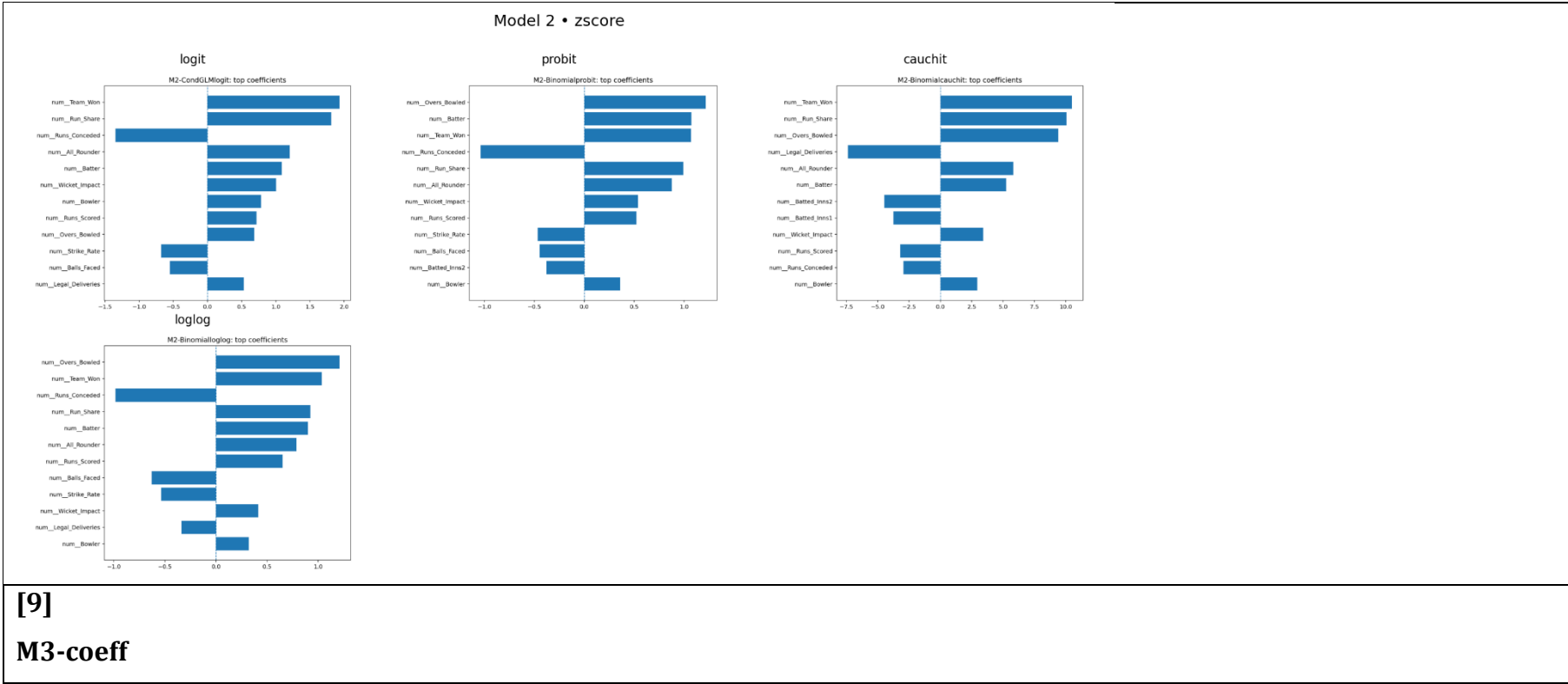
M2- Coeff

Model 2 • raw

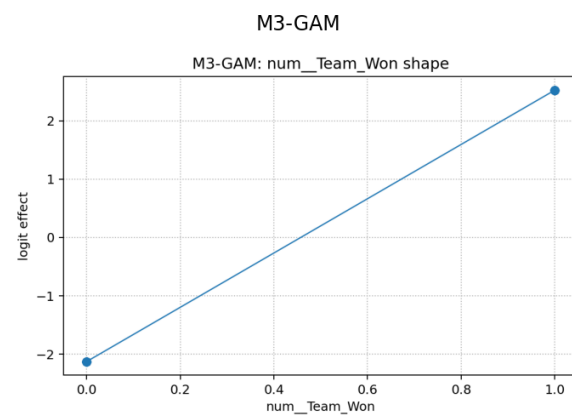


Model 2 • scaled

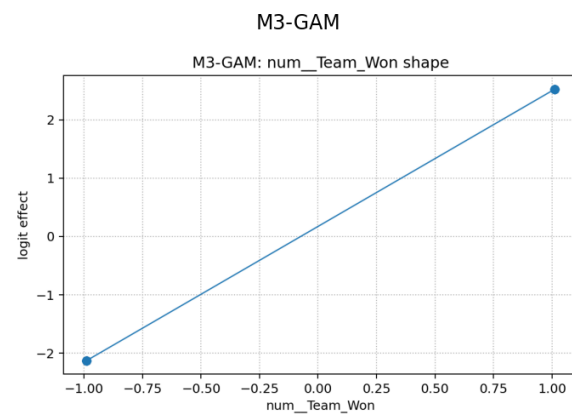




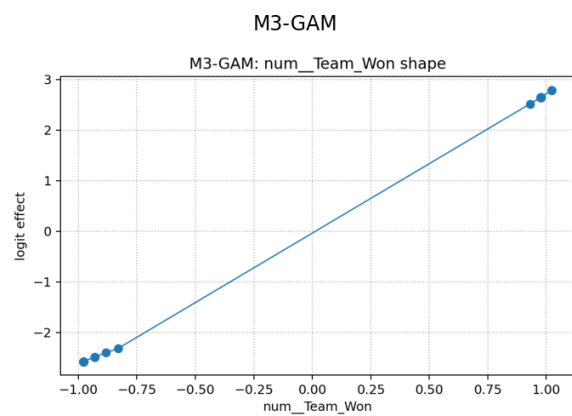
Model 3 (GAM) • raw



Model 3 (GAM) • scaled

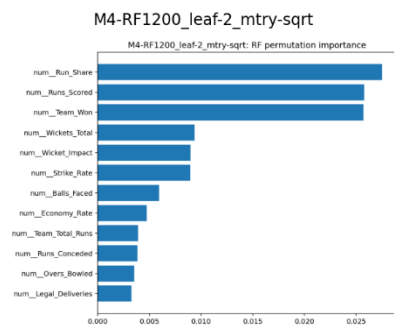


Model 3 (GAM) • zscore

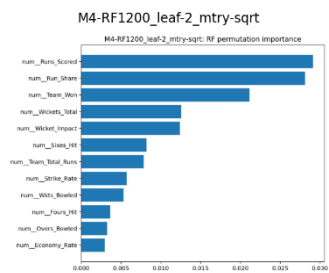


[10]

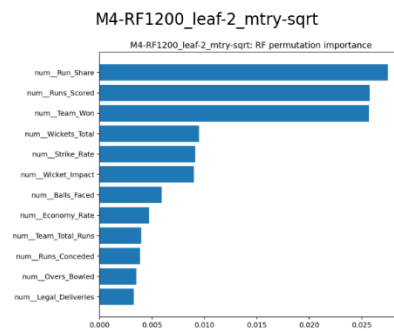
Model 4 (RF) • raw



Model 4 (RF) • zscore

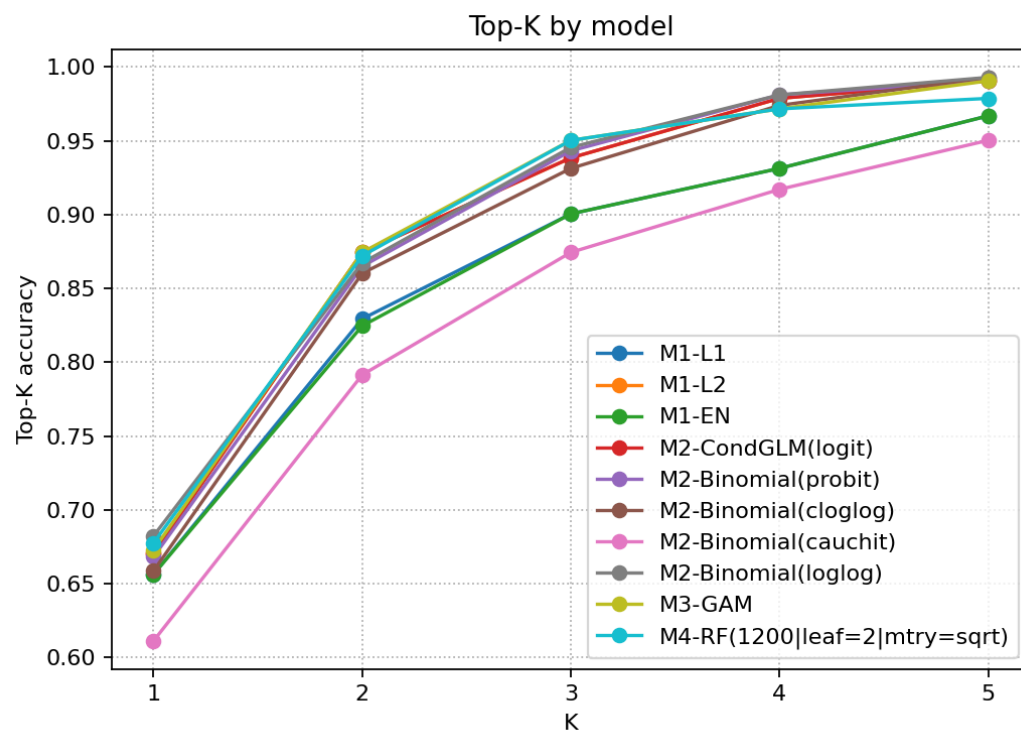


Model 4 (RF) • scaled



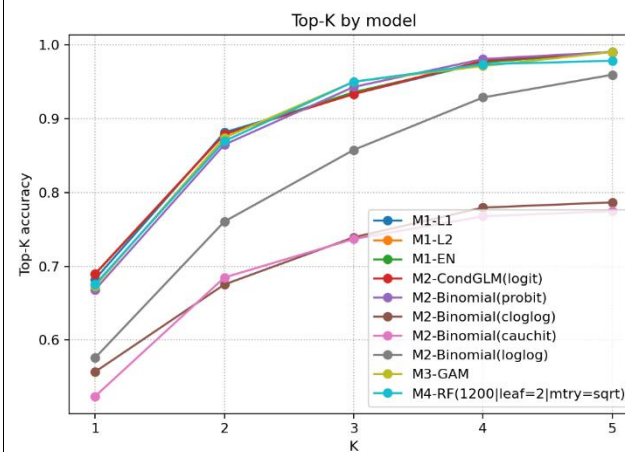
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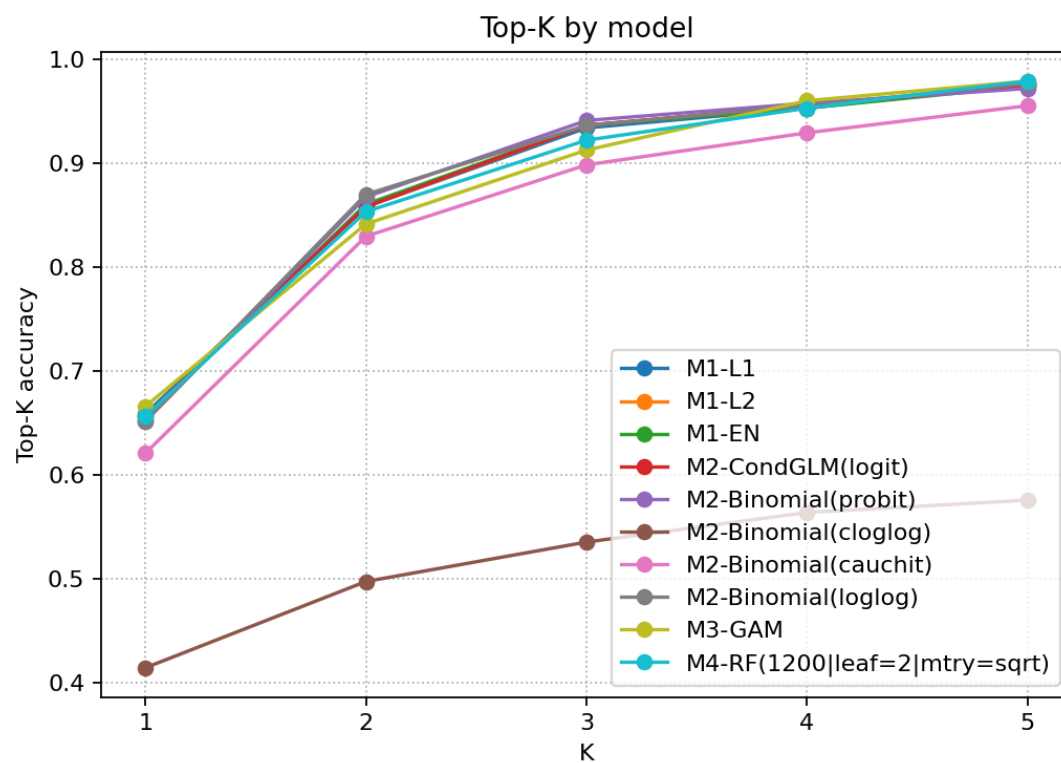
top - k - raw



[12]

Top-k global features

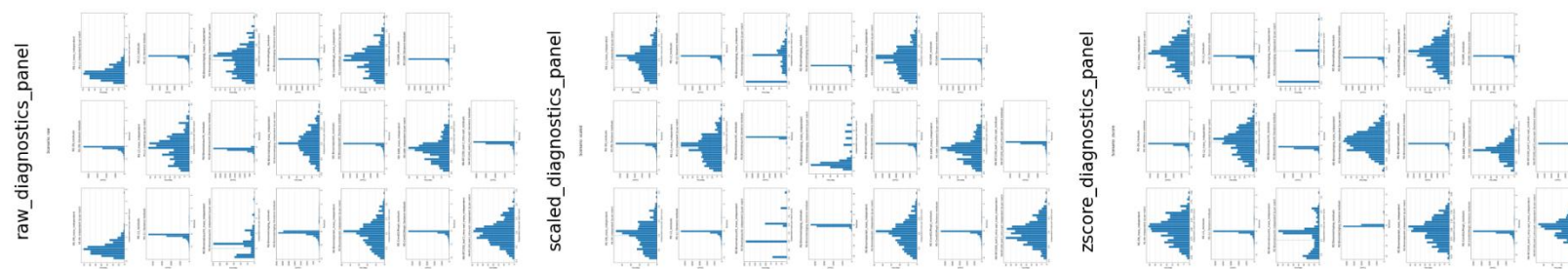


[13] top-k Z0score

[15]

residuals

Residuals + Mass Diagnostics Across Scenarios

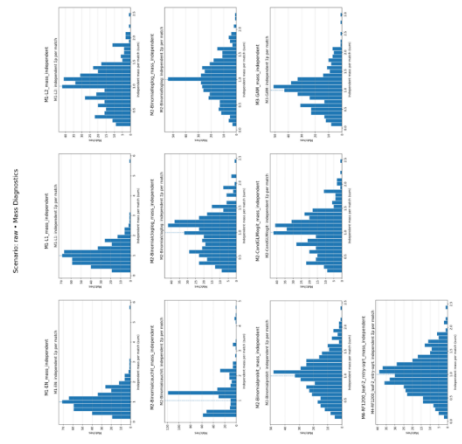


[16]

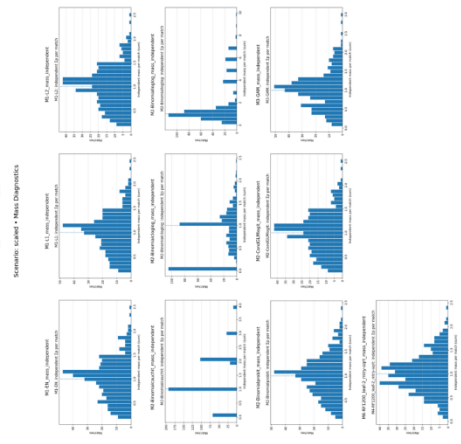
Mass charts

Mass Diagnostics Across Scenarios

raw_mass_panel



scaled_mass_panel



zscore_mass_panel

